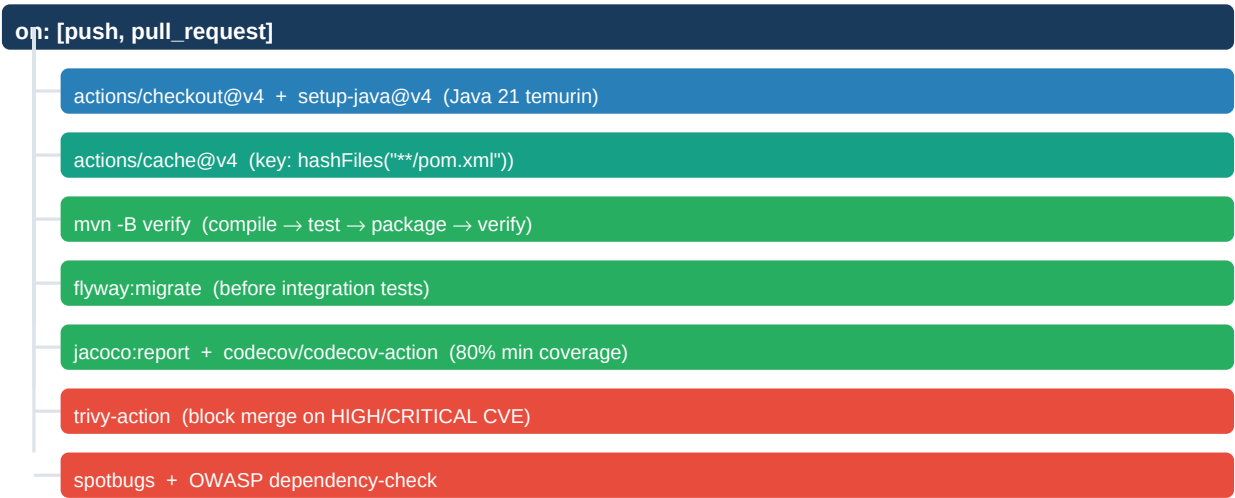


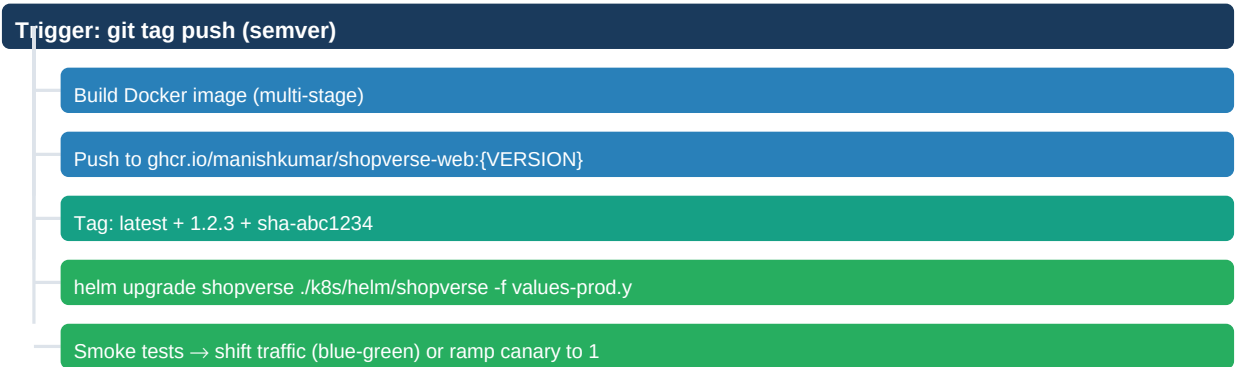
Overview

GitHub Actions CI/CD, Terraform IaC for AWS and GCP, OpenTelemetry observability, blue-green and canary deployments, and shift-left DevSecOps with trivy + SpotBugs + OWASP dep-check.

CI Pipeline (.github/workflows/ci.yml)



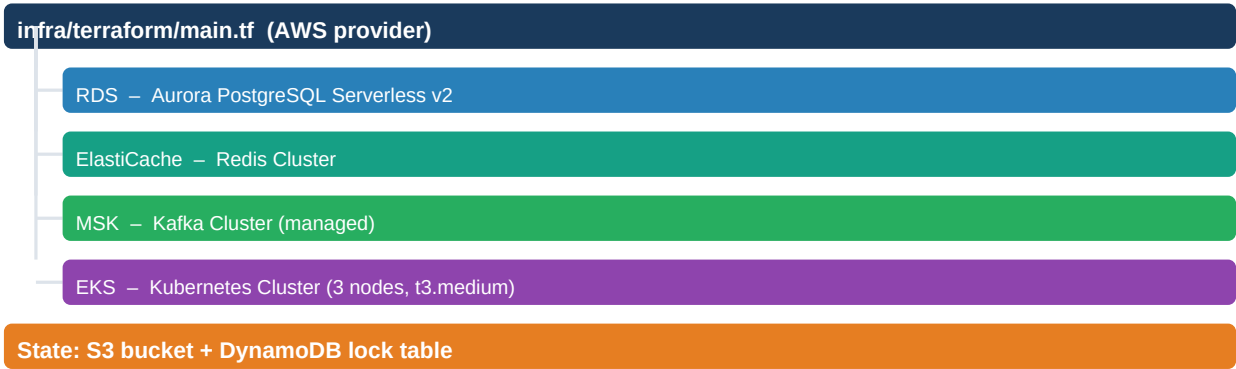
Release Pipeline (ci/release.yml)



Deployment Strategies

Strategy	Mechanism	Rollback	Risk
Rolling Update	K8s RollingUpdate (default)	Automatic on probe fail	Low, slow
Blue-Green	Deploy shopverse-green; switch Nginx upstream (flip switch)	Instant (flip switch)	Low
Canary	Nginx split_clients 10% → shopverse-canary; deploy 0.5%	Deploy canary; deploy 0.5%	Very low

Terraform IaC (AWS)



AWS → GCP Service Equivalents

Component	AWS	GCP
Relational DB	RDS Aurora PostgreSQL	Cloud SQL (PostgreSQL)
Cache	ElastiCache Redis	Memorystore Redis
Kafka	MSK (Managed Streaming)	Pub/Sub
Object Storage	S3	Google Cloud Storage
Container Orchestration	EKS	GKE
Document DB	DocumentDB (MongoDB API)	Firestore
Serverless	Lambda + API Gateway	Cloud Run + Cloud Functions